

Fluctuation Law: κ Dependence Test

$Q = \sigma_N/\sqrt{N}$ Depends on κ Through $T_{\text{avg}}(\kappa)$

The Law Is Valid; Its Scope Requires Clarification

Tests CRF_FluctuationLaw Priority 2. V20.5 · $N = 500$ · $\kappa \in \{0.05, 0.15\}$

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Abstract

CRF_FluctuationLaw states $Q \equiv \sigma_N/\sqrt{N} = 1/\sqrt{n \cdot T_{\text{avg}}}$ and predicts this is κ -independent (Priority 2 test).

Result: Q is not κ -independent when $T_{\text{avg}} = 35.42$ is held fixed. At $\kappa = 0.05$: $Q_{\text{measured}} = 0.235$ vs $Q_{\text{pred}}(T = 35.4) = 0.059$ — a 300% discrepancy. The measured T_{avg} at $\kappa = 0.05$ is 52.1 sweeps, not 35.4.

Root cause: In V20.5, the firing threshold is $\theta_{\text{eff}} = K^* \cdot \sqrt{M_i}$ where $M_i = 1 + \kappa N_{\text{eff},i}$. Smaller κ means slower mass growth, slower threshold rise, and a *different* $T_{\text{avg}}(\kappa)$. The Wald derivation (#21a) was performed at $\kappa = 0.15$; it does not give $T_{\text{avg}} = 35.42$ at other κ values.

Revised statement of the law:

$$\frac{\sigma_N}{\sqrt{N}} = \frac{1}{\sqrt{n \cdot T_{\text{avg}}(\kappa)}}$$

where $T_{\text{avg}}(\kappa)$ must be derived or measured for each κ . The law is structurally correct and connects to the δ -chain; its numerical value is κ -dependent through the mass-threshold coupling.

Check: $Q \cdot \sqrt{n \cdot T_{\text{avg}}(\kappa)} = 0.235 \cdot \sqrt{8 \times 52.1} = 4.80 \neq 1.0$, indicating that the functional form also requires revision — the law is not simply $1/\sqrt{nT_{\text{avg}}}$ with measured T_{avg} .

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1 Background

CRF_FluctuationLaw derived:

$$Q \equiv \frac{\sigma_N}{\sqrt{\bar{N}}} = \frac{1}{\sqrt{n \cdot T_{\text{avg}}}}$$

where $\sigma_N = \sigma_\Omega/\kappa$, $\bar{N} = \bar{\Omega}/\kappa$, $n = 8$ (SO(3)), and $T_{\text{avg}} = 35.42$ sweeps (#21a). The paper confirmed $Q = 0.05880 \pm 0.00040$ across 5 seeds at $\kappa = 0.15$, vs prediction 0.05941 (gap 1.0%).

Priority 2 test: does Q remain at 0.059 when κ is varied?

2 The κ Dependence: Root Cause

T_{avg} Depends on κ in V20.5

The V20.5 firing threshold is:

$$\theta_{\text{eff},i} = K^* \cdot \sqrt{M_i} = K^* \cdot \sqrt{1 + \kappa \cdot N_{\text{eff},i}}$$

At $\kappa = 0.15$: mass grows at rate 0.15 per event. A node at $\bar{N}_{\text{eff}} = 10$ events has $M = 1 + 0.15 \times 10 = 2.5$, so $\theta_{\text{eff}} = K^* \cdot \sqrt{2.5}$.

At $\kappa = 0.05$: the same node has $M = 1 + 0.05 \times 10 = 1.5$, so $\theta_{\text{eff}} = K^* \cdot \sqrt{1.5}$ — a lower threshold.

Lower threshold $\Rightarrow$ nodes fire more easily $\Rightarrow$ shorter T_{avg} ... but at the same time, nodes need *more* events to accumulate the same Ω , so $\bar{N}_{\text{eff}}$ is larger.

The net effect on T_{avg} depends on which effect dominates. **Measurement:** $T_{\text{avg}}(\kappa = 0.05) = 52.1$ sweeps $>$ $T_{\text{avg}}(\kappa = 0.15) = 35.4$ sweeps. The threshold growth effect dominates: smaller κ gives *longer* T_{avg} .

The Wald derivation (#21a) computed:

$$T_{\text{avg}} = \frac{2nK^*}{(n-1)\varepsilon_0} = 35.42$$

using the *initial* threshold K^* (before mass accumulation). This is valid at $\kappa = 0.15$ where mass effects are moderate. At $\kappa = 0.05$, the threshold-mass coupling changes T_{avg} significantly.

3 Measurement at $\kappa = 0.05$

Quantity	$\kappa = 0.15$	$\kappa = 0.05$	Note
T_{avg} (measured)	35.4 sw	52.1 sw	$\kappa = 0.05$ fires more slowly
$\bar{N}_{\text{eff}}$ (at 800 sw)	~ 10	11.5	more events but lower mass
$\bar{\Omega}$ (at 800 sw)	~ 1.5	0.57	lower: κ small
Q_{measured}	0.05880	0.235	$\times 4$ larger
$Q_{\text{pred}}(T = 35.4)$	0.05941	0.05941	unchanged
Gap	1.0%	$> 300\%$	—
$Q \cdot \sqrt{nT_{\text{avg}}^{\text{meas}}}$	0.059 $\sqrt{8 \times 35.4}$ 0.99	$\times$ 0.235 $= \sqrt{8 \times 52.1}$ 4.80	$\times$ should be 1.0

The product $Q \cdot \sqrt{nT_{\text{avg}}^{\text{meas}}} = 4.80 \neq 1.0$ shows that even with the measured T_{avg} , the functional form $Q = 1/\sqrt{nT_{\text{avg}}}$ does not hold at $\kappa = 0.05$. The law requires revision beyond simply using the κ -dependent T_{avg} .

4 What the Original Law Does and Does Not Claim

The Original Law Is Valid at $\kappa = 0.15$

CRF.FluctuationLaw confirmed $Q = 0.05880 \pm 0.00040$ across 5 seeds at $\kappa = 0.15$, matching $1/\sqrt{nT_{\text{avg}}} = 0.05941$ at gap 1.0%.

This result is valid. What it claims:

- Q is a stable, reproducible invariant at fixed κ .
- Q is connected to n and T_{avg} via the δ -chain at the canonical $\kappa = 0.15$.
- Variance suppression factor $\sqrt{nT_{\text{avg}}} \approx 17$ is confirmed at $\kappa = 0.15$.

What the original law does *not* establish:

- That Q is κ -independent.
- That T_{avg} is κ -independent.
- The functional form of $Q(\kappa)$ for $\kappa \neq 0.15$.

5 Status Table

Result	Status	Note
$Q = 0.059$ at $\kappa = 0.15$, 5 seeds	Confirmed	FluctuationLaw original
$T_{\text{avg}}(\kappa = 0.05) = 52.1 \neq 35.4$	Confirmed	measured V20.5
$Q(\kappa = 0.05) = 0.235 \neq 0.059$	Confirmed	κ dependence real
Q kappa-independent (prediction)	Falsified	Priority 2 test fails
$Q = 1/\sqrt{nT_{\text{avg}}(\kappa)}$ with meas. T	Falsified	product = $4.80 \neq 1$
Correct functional form $Q(\kappa)$	Open	requires derivation
$T_{\text{avg}}(\kappa)$ derivation	Open	Wald formula at arbitrary κ
Law valid at $\kappa = 0.15$	Confirmed	scope clarified

6 Open Questions

Next Targets

1. $T_{\text{avg}}(\kappa)$ **derivation (Priority 1)**. Extend the Wald derivation (#21a) to include the mass-threshold coupling $\theta_{\text{eff}} = K^* \sqrt{1 + \kappa N_{\text{eff}}}$. This requires a self-consistent equation since N_{eff} itself depends on T_{avg} .
2. $Q(\kappa)$ **functional form (Priority 1)**. Run $\kappa \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ at longer runs (1500 sweeps) and fit Q vs κ . If $Q \propto \kappa^\alpha$: identify α and derive it from the δ -chain.
3. **Re-examine collapse premium connection (Priority 2)**. The connection $\sqrt{nT_{\text{avg}}} \approx \beta/\delta\bar{H} \approx 17\text{--}19$ was identified at $\kappa = 0.15$. Does this ratio remain near 17–19 at other κ ? If yes: the connection is κ -independent even if Q is not.

The law held at $\kappa = 0.15$.

At $\kappa = 0.05$, it did not.

The boundary of a law is as informative as the law itself.

The mass-threshold coupling is where κ enters.

That coupling is what needs to be derived next.